

Wrocław University of Science and Technology
Faculty of Management

Business simulations

DES Project - Journey to Mordor

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1 Project details

1.1 Description

This project involves building a discrete event simulation model to represent a Middle-earth service facility named "Journey to Mordor Inc.", which is responsible for preparing four types of heroes:

- Hobbits
- Dwarves
- Elves
- Humans

for an upcoming great battle. As the heroes arrive at the facility in batches of ten, they must navigate a specific sequence through various service stations, such as a Weapon Sharpening Station (WSS), an Elven Hair (ELS), Gandalf Consulting (GCO), Morale Boosting (MBO) and a Lembas Snack Stop (LSS). Movement between these preparation stations is managed by a logistical transport system utilizing two magical M.U.L.E.'s that can carry one hero at a time. The fully animated simulation will track key performance measures like waiting times, total transport times, and station utilization over multiple replications, providing a baseline model that will be used to propose and test creative improvements aimed at reducing the heroes' total waiting time in the system.

1.2 Project requirements

All of the requirements from the task's text are as follows:

- Specific Material Handling blocks such as Station, Request, Transport and Free must be utilized
- Each hero type must be represented by a different Entity Picture
- An *Entity.sequence* must be defined based on the hero type
- A *Hero.type* attribute must be defined for each race (with different values)
- Simulation setup - 5 replications, each consisting of 16 hours per day with minutes as a base time unit
- Display of variables - such as current number of each hero type in the process, a countdown timer for the Great Battle at the Gates of Mordor and at least one additional important variable - is required
- All heroes must follow their predestined path up until meeting everyone else at the Gates of Mordor

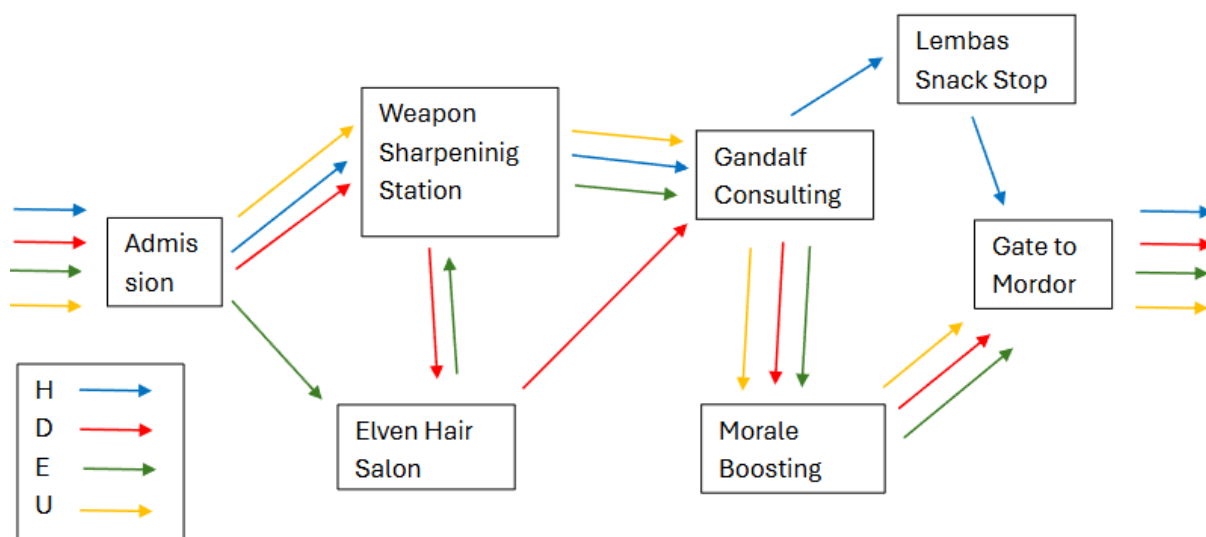


Figure 1: Scheme of preparation sequence for four types of heroes

2 Base model

2.1 Model overview

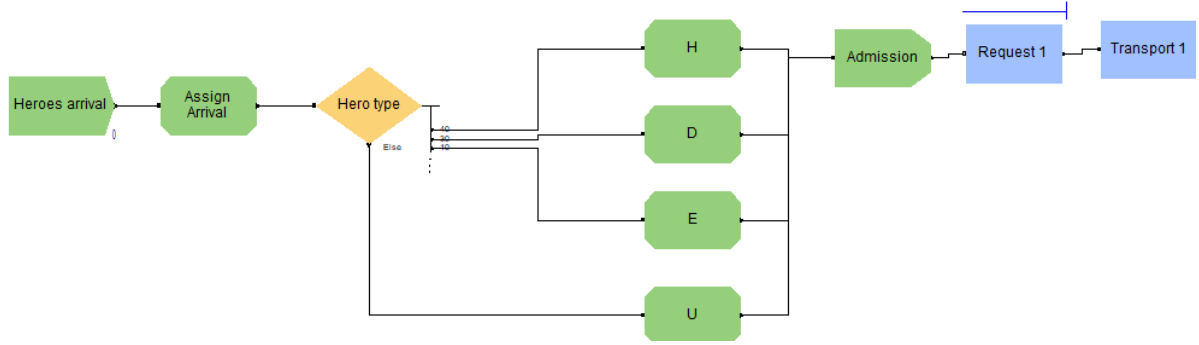


Figure 2: First part of the base model

The first part of the model focuses on assigning proper attributes to each hero arriving on the battlefield based on a principle of randomness (because life is random): 40% of heroes are Hobbits, 30% are Dwarves, 10% are Elves, and the rest are Humans. After that, all of the heroes sign up in the Admission station, where they request to be transported by M.U.L.E. to their destined location.

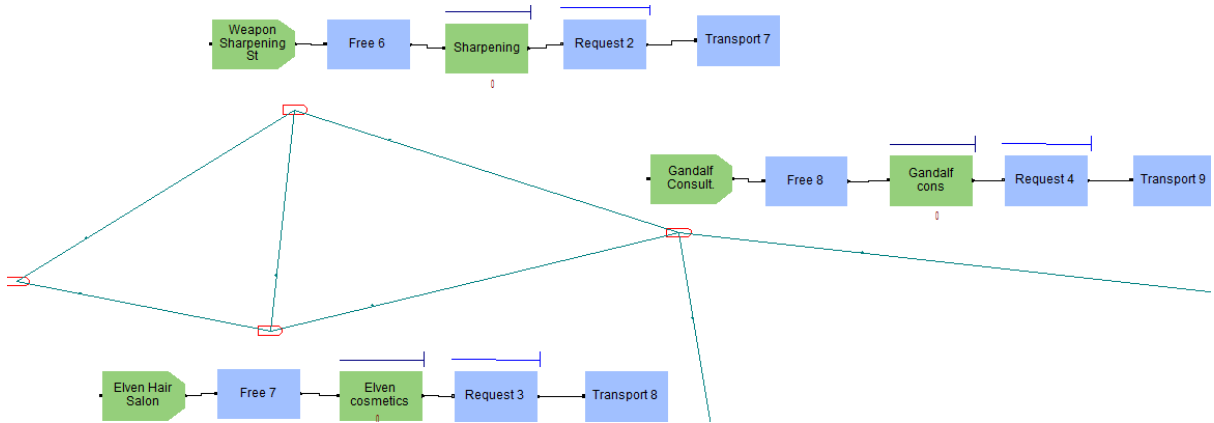


Figure 3: Second part of the base model

The second part of the model showcases three different stations that heroes need to stop by:

- the Weapon Sharpening Station, where a working Blacksmith can be found
- the Elven Hair Salon, occupied by elvish stylists
- and the Gandalf Consulting Station, where the old and wise wizard shares his knowledge.

Each path is similar, yet unique in its own way, requiring a different amount of time to be spent.

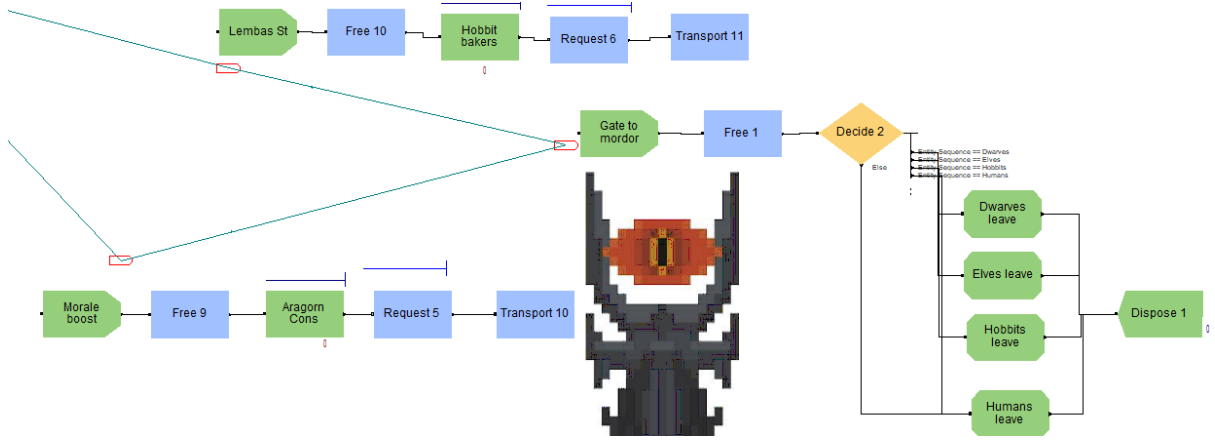


Figure 4: Third part of the base model

The third and last part of the base model includes two more stations - Lembas Snack Stop, where hobbit bakers prepare their precious meals, and Morale Boosting Station, where Aragorn breathes courage into our heroes. In the end, everyone meets each other at the Gates of Mordor, where all forces assemble and charge to deal with Sauron once and for all. As the entities leave our simulation, they also pass additional "Assign" blocks, which help us track the hero count throughout the whole simulation.

2.2 Heroes representations inside simulation



Figure 5: Hobbit's image



Figure 6: Elf's image



Figure 7: Dwarf's image



Figure 8: Human's image

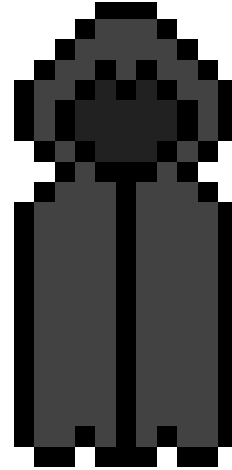


Figure 9: Mystery man's image - until a hero is born

2.3 M.U.L.E.'s representation inside simulation



Figure 10: First M.U.L.E.



Figure 11: Second M.U.L.E.

2.4 Results of the base model

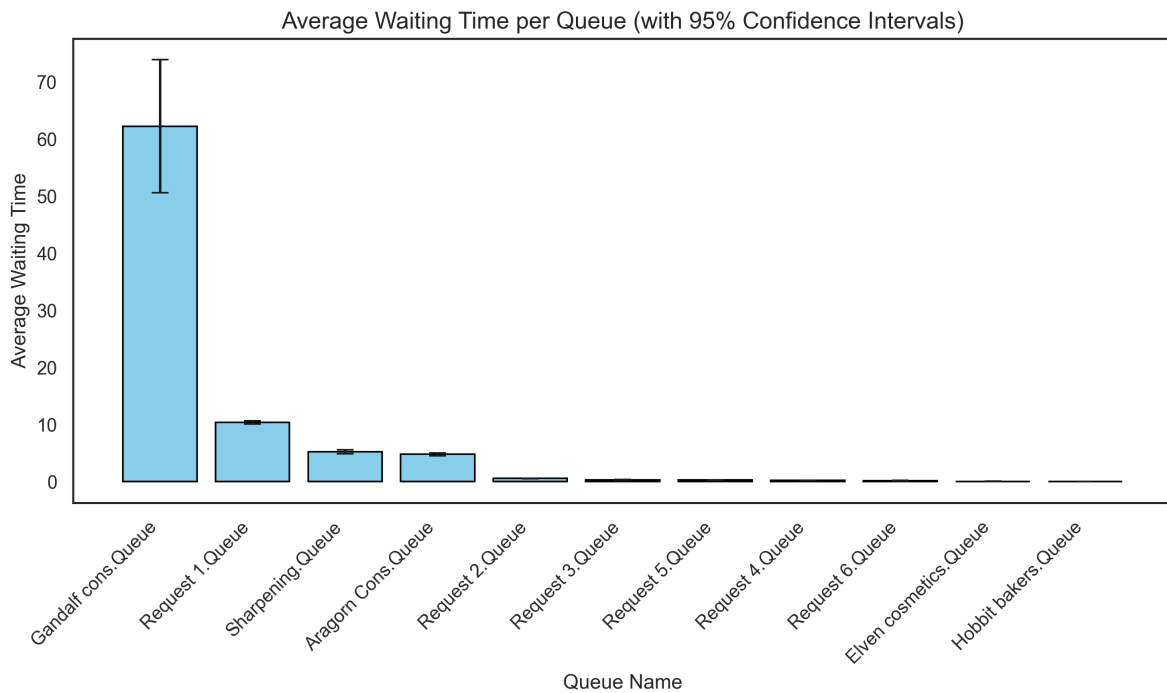


Figure 12: Average waiting time per queue for the base model

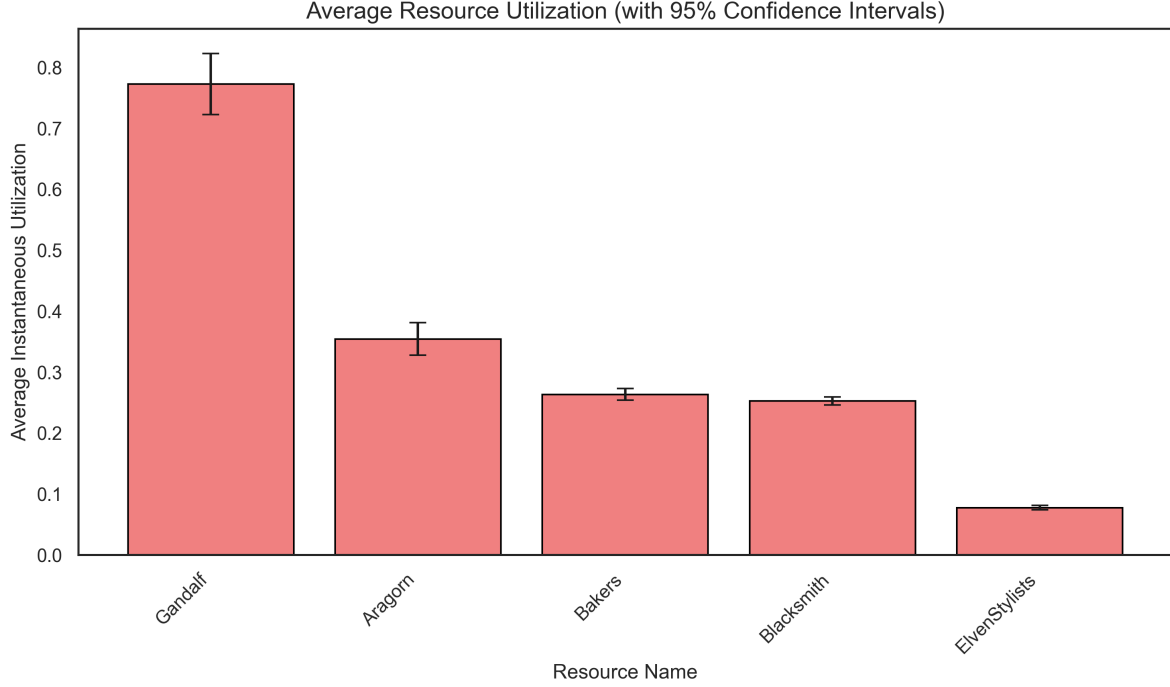


Figure 13: Average resource utilization for the base model

Selected metrics for the evaluation of the basic model were as follows:

- Preparation time at each station
- Delays for each station
- Queues for M.U.L.E.'s - especially for transport modules
- Minimum and maximum transport time along with it's average value
- Utilization of all resources and M.U.L.E.'s

The preparation time varies significantly at each workstation. At Aragorn's, there is an 11 to 12 minute preparation time. At elven cosmetics, it's also about 11 minutes. The time changes at Gandalf's, where we need 14 minutes. At Hobbit's bakery, about 25 minutes are needed, and lastly, at the Blacksmith's weapon sharpening station, where there is around 5 minutes of preparation time. On average, every hero needs 55 minutes to prepare for the great battle and travel to Mordor. When analyzing the queues for transport modules and M.U.L.E.'s, the most important request is the first one happening at the admission station, where heroes need to wait for 10 minutes on average. At the other stations, there is barely a queue, and they don't need to wait more than one minute. The transfer time of a single hero is 5.67 minutes on average. The intervals of transport time between stations are between 3 and 7 minutes.

Gandalf and Aragorn were the hardest working people - during the simulation, their utilization values were 77% and 35%. Bakers and blacksmiths share slightly less but similar values, oscillating around 25%-26%. The elves do not have much to do due to their 7% utilization value. M.U.L.E.'s. were utilized for 29% of the simulation's time. The most used queues were on admission, where 54% of the time spent was essentially waiting, and at the Sharpening station, holding a value of 27%.

2.5 Proposed modifications

To conclude the results of the base model, the longest queues form at the beginning of the hero's journey, at the starting station. Gandalf and Aragorn were the most frequently used resources, and the M.U.L.E. vehicles do not cause the system to become overloaded. The real challenge for the modified model is managing the capacity of Aragorn and Gandalf, the speed of the M.U.L.E. vehicles, or adding another transporter to reduce queues at the admission.

3 Modified model

3.1 Biggest changes

Below are listed all of the changes applied to the base version of the model to increase it's effectiveness:

- Increased the number of M.U.L.E.'s available to four, effectively doubling their initial number, thanks to Theoden and the kingdom of Rohan, who gave their own horses — Hasufel and Snowmane



Figure 14: Third M.U.L.E.



Figure 15: Fourth M.U.L.E.

- Increased M.U.L.E.'s initial speed by 50%, thanks to giving heroes riding them a stick with a carrot, feeding them Lembas bread prepared by Hobbit bakers, and changing their battle armor to the mithril one
- Increased Gandalf's efficiency by giving him enough pipe-weed from the Shire and Miruvor to bear everyone's troubles on his shoulders. In this state, he can guide three heroes at once!
- Increased the number of blacksmiths by one - elves made peace with dwarves and have sent one smith from Rivendell (though reluctantly)

3.2 Results of the modified model

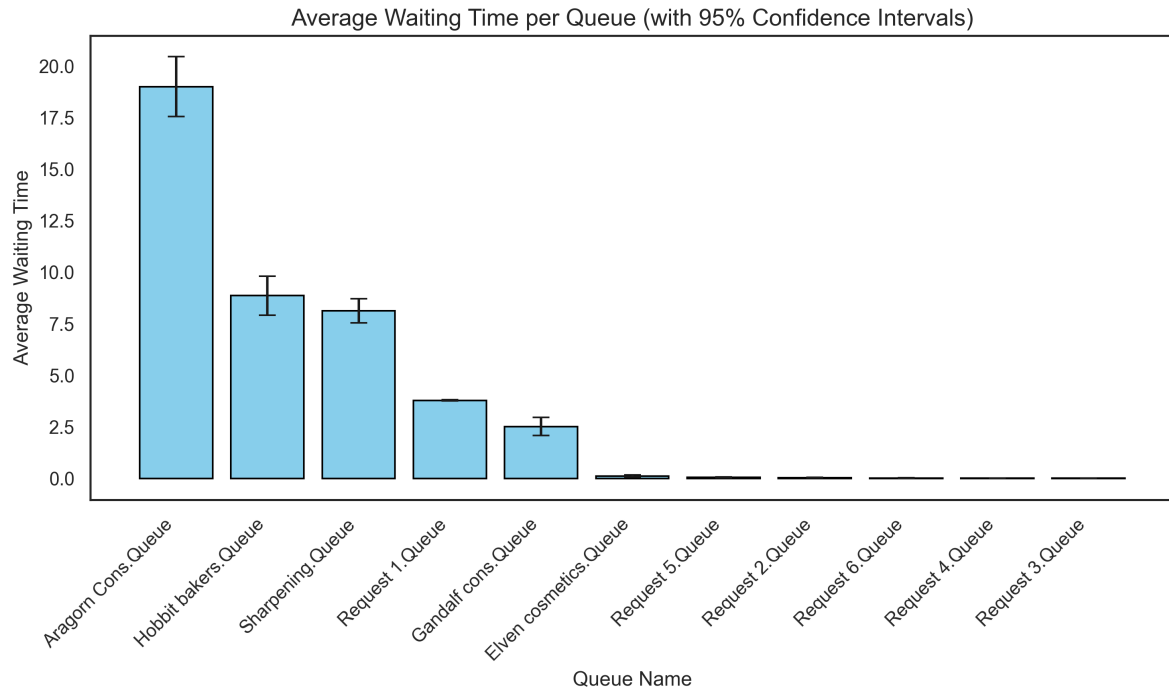


Figure 16: Average waiting time per queue for the modified model

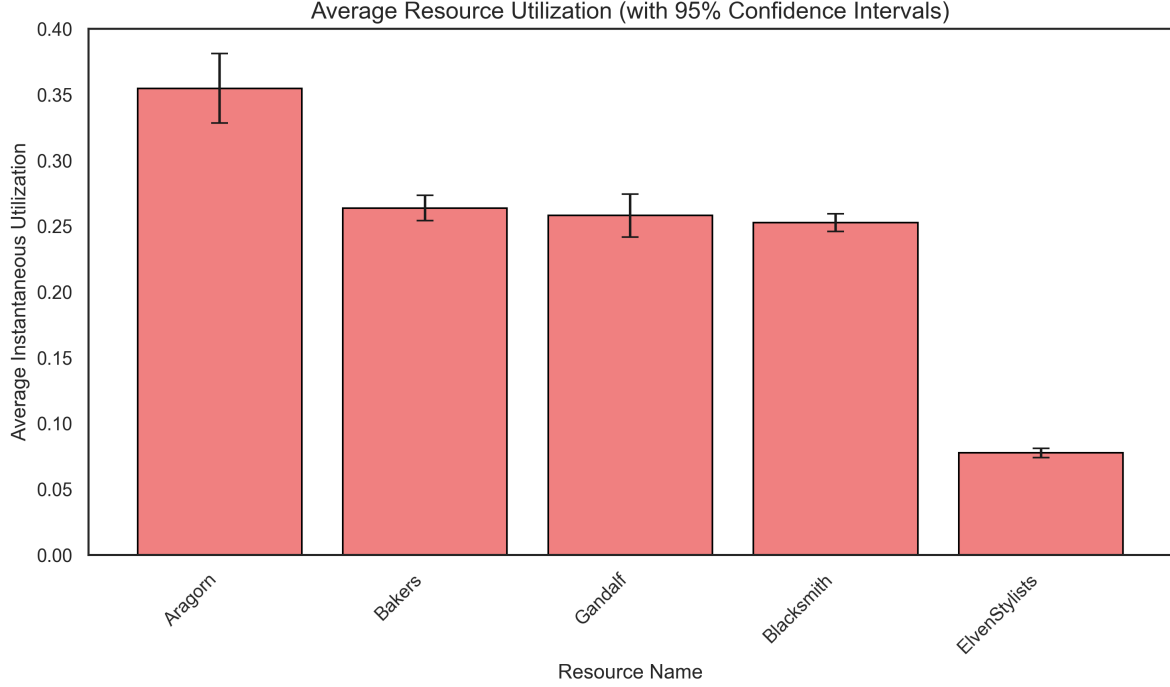


Figure 17: Average resource utilization for the modified model

The modifications — specifically increasing Gandalf’s guiding capacity, adding another blacksmith, and upgrading the M.U.L.E.’s fleet — have been highly effective in their targeted areas, successfully reducing the resource utilization for both Gandalf and the blacksmiths to a highly manageable value of around 25%, while keeping the time spent in queue for their expertise relatively low. However, the overall condition of the simulation reveals that the system remains unbalanced because the bottlenecks have simply shifted to different places rather than ceasing to exist. For now, the biggest congestion is now located at Aragorn’s, who suffers from the highest average wait time in the system at 19 minutes, followed closely by the Hobbit bakers’ queue, standing at approximately 9 minutes. Ultimately, while the Rohan, Shire, and Rivendell sponsored upgrades provided massive relief to some degree, the simulation will not work as smoothly as it could until Aragorn’s case is solved.

4 Note on project members contribution and LLM use

Hubert was mainly creating the first part of the base model, which consisted of creating adequate "Assign" blocks and defining hero-types based on a "Decide" block. The other thing he focused on was adding the appropriate entity pictures (M.U.L.E.’s, Hobbits etc.), as well as the Sauron’s eye. His contribution also includes creating additional experiments within the modified model, as well as adding timers for the great battle, which count every entity’s presence inside the simulation.

Bartosz focused primarily on creating all of the stations and connecting them with routes. "Transporters", "Distances" and the logic between them were established within the "Sequences" and "Distances" module. The final steps involved compiling the results of the base model and preparing a conclusion.

Despite having different tasks and implementing distinct parts of the model, we collaborated throughout the entire project and helped each other by spotting any errors. Debugging was a common occurrence, and we worked together to resolve these issues. LLM models were not used at any stage of the task. AI engines lack a sufficient understanding of Arena Software’s framework, which means that they generate structures that do not reflect our situation. We relied solely on our knowledge of Arena, the materials sent beforehand, and the tips provided via e-mail and on the e-portal.